

Multiphysics modelling and preliminary analysis on the freeze valve behaviour in a Molten Salt Fast Reactor¹

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Abstract

In the Molten Salt Fast Reactor (MSFR), the safety function related to criticality and cooling is also delivered via draining tanks developed to accommodate liquid fuel discharged by gravity during normal maintenance or accidental events. In particular, a safety barrier that is under investigation to manage accidental scenarios consists of a salt-frozen plug designed to melt in case of loss of power or overheating. Specifically, the salt plug should melt as a consequence of an unintended temperature increase, allowing the salt to drain into a safety tank. The mechanism based on the state of the plug embraces the passive safety concept, highlighting the importance of delving into the steady state of the freeze valve and the transient behaviour. This paper proposes a preliminary analysis of the behaviour of the freeze valve within a domain that represents a symmetric portion of the MSFR primary loop. The model, developed in OpenFOAM, couples melting and solidification phenomena to a multiphysics solver, coupling neutronics and thermal-hydraulics. The 3D domain represents 1/16th of the MSFR fuel loop and is equipped with a cylindrical region that mimics the presence of the freeze valve. Two scenarios are considered to assess the impact on the freeze plug behaviour of initial and boundary conditions meant to represent different cooling strategies. When the plug is coupled with the MSFR fuel loop, the selection of boundary and initial conditions strongly affects the plug melting time. This highlights the need for design specifications for the geometry and the cooling mechanisms to correctly operate the freeze valves.

Keywords: Molten Salt Fast Reactor; Freeze valves; Multiphysics model; OpenFOAM

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1 Introduction

The Molten Salt Fast Reactor (MSFR) is the reference concept chosen by Generation IV International Forum as a representative of circulating fuel reactors (Forum, 2002). As currently conceived for the MSFR, the safety functions related to criticality and cooling are also delivered through draining tanks located below the core, meant to accommodate the liquid fuel, discharged by gravity. These tanks are developed to operate in both normal maintenance and during accidental scenarios, such as active cooled temporary storages or criticality-safe tanks, respectively. The latter are part of the emergency drainage system, whose opening devices show great potential to satisfy the passive safety objective. The working principle lays in the melting of freeze plugs in case of loss of power or overheating, ensuring a fast draining of the circulating fuel salt in the safety tanks. The freeze plugs should be positioned in a portion of the draining pipes below the reactor and they have the great advantage of compatibility to the harsh MSFR environment.

In the context of European projects as SAMOFAR (SAMOFAR) and SAMOSAFAER (SAMOSAFAER), as well as at the Shanghai Institute of Applied Physics and at the University of Electro Communications in Chofu-shi the analysis of the freeze plugs and the emergency drainage system was pursued by performing experimental campaigns and numerical simulations (Tiberga et al., 2019; Giraud et al., 2019; Tano et al., 2017, 2018; Li et al., 2014; Aji et al., 2018). Actually, the idea of exploiting frozen salt plugs designed to melt allowing for the salt drainage belongs to the Molten Salt Reactor Experiment (MSRE) developed at the Oak Ridge National Laboratory (ORNL) (Richardson, 1962). Anyway, this mechanism relied on active systems as heaters and coolers to control the state and operational mode of the valve. In this study, the dependence of the valve state and its progressive opening on the temperature evolution aligns with the concept of passive safety. The solid plugs of salt are expected to melt in the case of an unintended temperature rise, allowing the discharge of the liquid fuel in emergency drainage system. This feature justifies the interest in providing numerical simulations the capability to implement, within the circulating fuel salt models, phase change phenomena. In this light, this work extends the research line followed at the Politecnico di Milano (Tartaglia et al., 2023) to develop a multiphysics solver coupling in the same environment thermal-hydraulics and neutronics, while also modelling the melting and solidification phenomena, through latent heat and velocity transition. It is worth mentioning that multiphysics analyses play a major role in assessing the behaviour of molten salt reactors, due to the strong coupling between neutronics and thermal-hydraulics. Significant research efforts are currently being pursued for the development and advancement of numerical multiphysics tools aimed at capturing the circulating fuel governing phenomena (Fiorina et al., 2015; Jaradat et al., 2024; Lee et al., 2025).

The tool proposed in this work is suitable to provide insights into melting phenomena such as the behaviour of the freeze valve, as well as investigating salt solidification in regions subjected

to the lowest temperature. For example, once the design specifications for the heat exchanger component are available, the solver could be used to assess the stratification of solid salt layers in different portions of the heat exchanger.

In this work, the ultimate objective is to develop a model to analyse the freeze valve behaviour at nominal operational conditions, currently under consideration as a safety barrier for the MSFR. The multiphysics simulations involve a 3D domain representing 1/16th of the MSFR primary circuit, equipped with a cylindrical region to mimic the presence of the freeze valve. A square cylinder of 20 cm diameter is added to the bottom of the cold leg. The goal is to verify whether this geometry allows the salt valve to withstand the harsh MSFR working conditions, ensuring a complete closure of the valve when coupling with the MSFR fuel loop. Furthermore, different initial and boundary conditions are imposed on the plug to represent the cooling strategies that could be exploited in combination with the freeze valve mechanism. Therefore, as a preliminary investigation, this study implements fixed value boundary conditions and external wall heat flux boundary conditions to simulate active cooling. Overall, the analyses focus on the dependence of the freeze plug state to different operating conditions and assess the flexibility of a multiphysics solver, developed in the OpenFOAM toolkit, coupling in the same framework thermal-hydraulics, neutronics and phase change phenomena.

The remainder of this paper is organised as follows: in Section 2, the multiphysics solver and the 3D model representing 1/16th of the MSFR are presented. In Section 3, a mesh sensitivity analysis is proposed aimed at investigating the freeze plug region to be included in the primary loop. Specifically, to lighten the computational burden, the simulations are performed on a square cylinder with the same dimensions as the portion that will be attached to the lower part of the reactor in the subsequent analyses. Finally, Section 4 encompasses a steady state analysis of the whole domain followed by two pseudo-transient scenarios where different initial and boundary conditions are imposed to the freeze plug domain. Some conclusions are then drawn concerning the freeze valve behaviour.

2 Numerical model

The goal of providing insights into the freeze valve behaviour requires the development of tools capable of solving the different physics involved in the same environment. This work follows the research efforts performed at the Politecnico di Milano and exploits two OpenFOAM solvers already developed at Polimi and comprehensively described in Tartaglia et al. (2023), Aufiero (2014), Di Ronco et al. (2021), Fiorina et al. (2022). The potentialities of the two solvers are merged in a unique tool coupling in the same OpenFOAM environment neutronics, thermal-hydraulics, melting and solidification phenomena. The resulting solver is adopted to perform multiphysics

simulations on a 3D domain representing 1/16th of the MSFR fuel loop, equipped in this work with a region mimicking the presence of the solid plug. In this section, the submodules composing the multiphysics solver are briefly presented, along with the MSFR simulation domain and characteristic properties. For the sake of clarity, the standard neutronics-thermal-hydraulics solver is first introduced, and then the modelling approach of phase change phenomena is described in the next subsection.

2.1 Multiphysics solver

The multiphysics solver presented in this work couples a neutronics-thermal-hydraulics solver, developed to perform transient simulations, with models to describe melting and solidification phenomena. Fig. 1 illustrates a schematic representation of the structure of the solver. Each main block contains a characteristic equation for a given variable, solved once at a time. In the standard version of the solver, the thermal-hydraulic module implements an incompressible single-phase version of the Reynolds-Averaged Navier–Stokes (RANS) equations, solving for pressure, velocity, and temperature of the liquid salt (Aufiero, 2014). The salt thermophysical properties are considered constant, and the Boussinesq approximation is employed to take into account buoyancy effects according to linearisation around a reference temperature T_{ref} , through a thermal expansion coefficient β in the momentum equation. For the neutronic modelling, the multigroup neutron diffusion approximation is adopted. Moreover, the toolkit provides the possibility to implement both vacuum and albedo boundary conditions, suitable to mimic the presence of upper and lower reflectors and of the fertile blanket. In addition, the solver implements transport equations for delayed neutrons and decay heat precursors to describe the drift of precursors induced by liquid fuel motion.

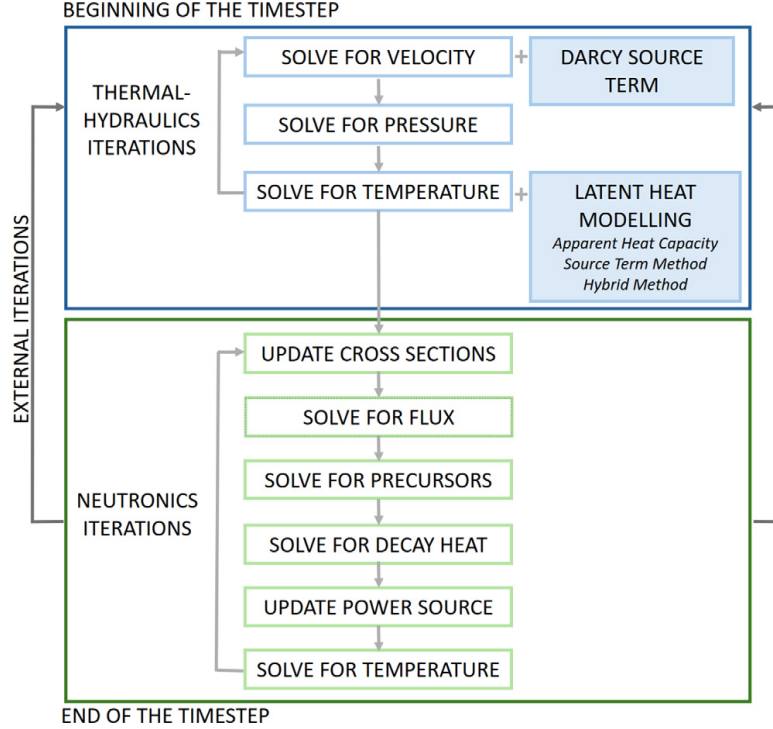


Fig. 1. Coupling strategy and structure of the solver, extending the scheme proposed in Tiberghia et al. (2020).

The coupling between these two modules is guaranteed by feeding the energy equation with a volumetric power source, accounting for both prompt fission and decay heat contributions. On the other hand, the neutronic group constants are updated according to temperature and density distributions to consider thermal feedback effects. The reference neutronic parameters are corrected through a linear term accounting for the thermal expansion related to density variations and a logarithmic term to represent the Doppler effect. To mimic the presence of the pump and the heat exchanger, a momentum source and a heat sink are included in the conservation equations. The parameters characterising the primary loop components can be modulated with appropriate time dependencies to simulate transient scenarios.

In line with the work accomplished at Politecnico di Milano (Tartaglia et al., 2023), melting and solidification phenomena are incorporated in the neutronics-thermal-hydraulics solver by modelling two main characterising aspects of phase change, namely latent heat and velocity transition. Since the phase transition is a multiscale and highly non-linear problem, the modelling choice lies on fixed grid models thanks to their simplicity and reasonable accuracy. Therefore, the melting interface is not reconstructed explicitly but rather smeared out in a region referred to as *mushy zone*, which accounts for the simultaneous presence of both the solid and liquid phases. This choice enables treating the latent heat under the continuum mixture theory, in which the two phases are considered separate subsystems and the mixture properties are determined by properly weighting the properties of each individual phase. Specifically, the solver allows to define dif-

ferent specific heat and density, considered temperature invariant, for the two phases and linking them to the properties of the mixture through the volume liquid fraction. To facilitate the coupling with the thermal-hydraulics-neutronics submodules, temperature-based models are selected, i.e., *Apparent Heat Capacity* and *Source Term Method*. For these simulations, a hybrid method taking advantages from the two aforementioned models is adopted. In particular, the latent heat model is switched according to a check performed after solving the energy equation. In particular, at the beginning of the timestep, the solution advances using the *Apparent Heat Capacity* method. Once the energy equation is solved, the updated temperature field is evaluated to determine whether the latent heat peak has been skipped, so whether any computational cell exhibits a temperature jump from below the solidus temperature ($T < T_s$) to above the liquidus temperature ($T > T_l$), or vice versa, without traversing the mushy zone. If a skip is detected, the temperature field is restored to its original value, and the model switches to the *Source Term Method*, iterating between the energy equation and the closure relation until convergence. The algorithm then proceeds to the next timestep. This approach leverages a good compromise between computational efficiency and robustness, preventing the undesirable issue of latent peak skipping. Furthermore, the non-linearities stemming from coupled variables in the energy equations are managed thanks to closure relations with fictitious continuous function to smear out the phase change in the *mushy zone*. In this case, a linear relation between temperature and volume liquid fraction, g , is imposed in the *mushy zone*, with a formulation presented in (1).

$$\begin{cases} g = 0 & \text{for } T < T_m - \varepsilon \\ g = \frac{T - T_m + \varepsilon}{2\varepsilon} & \text{for } -\varepsilon < T - T_m < \varepsilon \\ g = 1 & \text{for } T > T_m + \varepsilon \end{cases} \quad (1)$$

where $2\varepsilon = dT$.

Finally, the velocity transition phenomenon arising from different responses to shear stresses in the solid and liquid phases is included thanks to the *Darcy Source Term* approach. The mechanical aspects are neglected, but the RANS equations are modified in order to enforce a null velocity in the solid phase while maintaining the standard formulation for the liquid phase. Specifically, the momentum equation is provided with a source, S , that overcomes all other terms for the solid phase, resulting in a null velocity field, while it drops to zero for the liquid phase. This term is presented in Eq. (2), where C is a constant ranging from 10^8 to 10^{10} , while ϵ prevents numerical divergence.

$$S = \frac{(1 - g)^2}{(g^3 + \epsilon)} C \nu \mathbf{u} \quad (2)$$

For further details about all the equations implemented in the model, the reader is referred to

Tartaglia et al. (2023).

2.2 MSFR model

The 3D domain representing the 1/16th of the MSFR is depicted in Fig. 2, where different zones constituting the primary loop components are highlighted. With respect to the geometry adopted in simulations testing the reactor behaviour in thermal-hydraulics-neutronics multiphysics simulations (Deanesi et al., 2023), a square cylinder of 20 cm diameter is added to the bottom of the domain to reproduce the freeze valve. Furthermore, the thermophysical properties of the salt are summarised in Table 1.

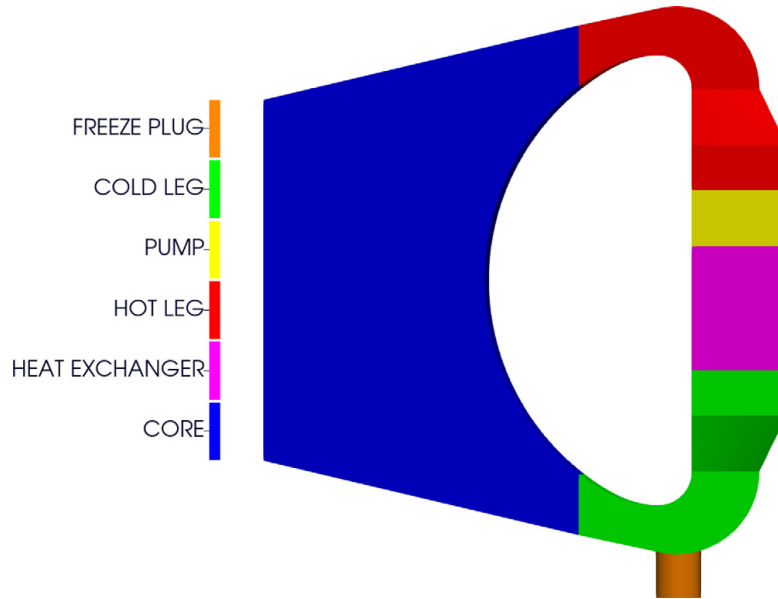


Fig. 2. OpenFOAM simulation 3D domain subdivided in zones for the primary loop.

Table 1. Reference thermophysical properties of molten salt.

Parameter	Symbol	Value	Units
Density	ρ	4306.71	kg m^{-3}
Specific Heat Capacity	c_p	1593.94	$\text{J kg}^{-1} \text{K}^{-1}$
Laminar Viscosity	ν	$5.89 \cdot 10^{-6}$	$\text{m}^2 \text{s}^{-1}$
Thermal Expansion Coefficient	β	$1.1912 \cdot 10^{-4}$	K^{-1}
Reference Temperature	T_{ref}	973	K
Intermediate Salt Temperature	T_{ext}	908	K
Melting Temperature	T_m	841	K
Latent Heat	L	$1.59 \cdot 10^5$	J kg^{-1}
Mushy zone width	dT	1	K

Due to the lack of design specifications at the current state of the art, the pump and the heat exchanger are represented as boxes where semi-implicit momentum and temperature sources are imposed. This modelling approximation is justified considering that these simulations are focused on evaluating the liquid salt behaviour in the core at the expense of simplifying assumptions on the description of the primary loop components. The momentum source and heat sink parameters are adjusted to meet target values for the main operational variables (Cammi et al., 2017). Concerning the heat exchanger, the volumetric heat sink can be expressed as:

$$q = \gamma (T_{ext} - T) \quad (3)$$

where γ ($\text{W/m}^3 \text{K}$) is a tunable parameter that accounts for the global heat transfer coefficient between the primary and intermediate loops and for an effective volume of the heat exchanger, while T_{ext} (K) is a reference temperature of the intermediate loop (Cammi et al., 2017).

For the neutronic part of the model, six energy groups are implemented for the fluxes in the multi-group diffusion approach, while eight and three groups are selected to represent the delayed neutron precursors and decay heat precursors, respectively (Aufiero, 2014; Fiorina et al., 2013). Vacuum boundary conditions are imposed on the walls corresponding to the reflectors and fertile blanket. The neutronic parameters are evaluated through a Monte Carlo simulation adopting the reactor physics and burnup code SERPENT 2 and the JEFF-3.1.1 cross section library. For further details the reader is referred to Cammi et al. (2017), Cervi (2019).

3 Mesh sensitivity analysis

In this work, with respect to the reference mesh representing 1/16th of the MSFR fuel loop, an additional region is attached to the lower part of the cold leg to reproduce the freeze plug. Therefore, as a first step, a preliminary sensitivity analysis is performed to inspect the generation of the mesh for the freeze plug region. To lighten the computational burden of the simulations, the mesh features are tested on a square cylinder with appropriate boundary conditions to simulate the coupling with the primary circuit. Since the focus is to estimate the impact of the refinement on the freeze plug domain, the analysis characterises the plug through a square cylinder with the same dimensions as the one included in the MSFR 1/16th domain. The height and the diameter of the cylinder are 20 cm. Accordingly, three meshes are built with different refinements, progressively doubling the number of hexahedral elements from a coarser to a finer mesh. The three meshes under analysis are depicted in Fig. 3, where (a) is the coarsest mesh, (b) has an intermediate level of refinement, and (c) is the finest one.

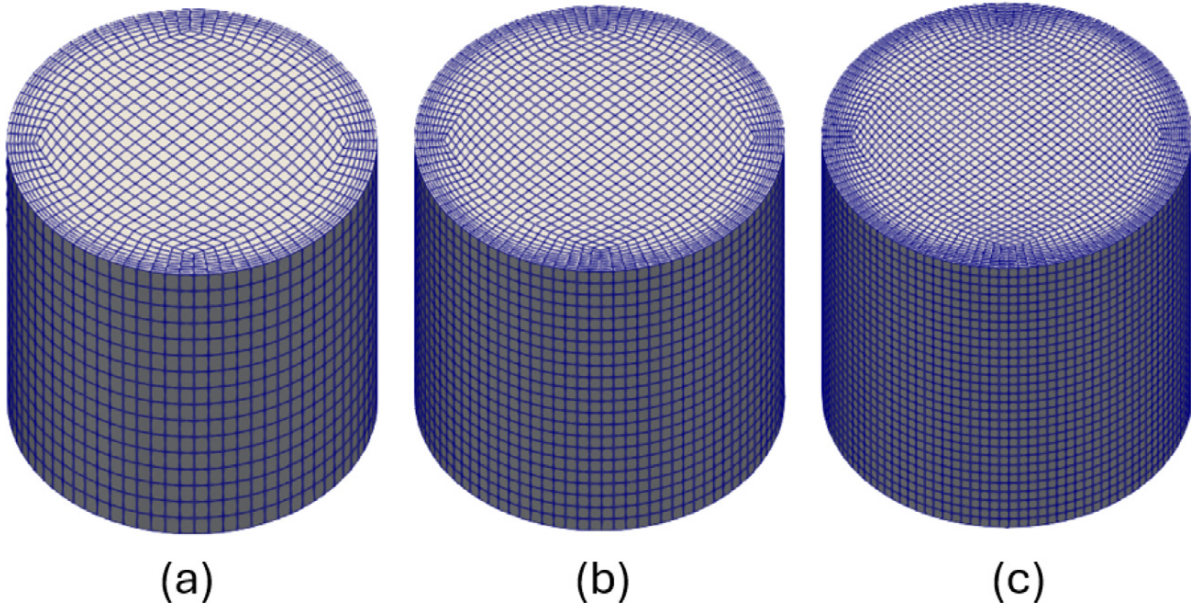


Fig. 3. Meshes adopted for the sensitivity analysis, with progressively finer refinement from left (a) to right (c).

To identify the influence of mesh refinement in that additional region, the Grid Convergence Index (GCI) (Roache, 1997) is employed as an indicator of the suitability of the mesh.

To evaluate the GCI, the approach proposed by Roache based on the Richardson Extrapolation is followed (Roache, 1998). Considering the three meshes where the subscript 1 coincides with the

finest grid notation, the GCI formulation reads:

$$GCI_{i+1,i} = F_s \frac{|\varepsilon_{i+1,i}|}{r^p - 1} \quad (4)$$

where F_s is equal to 1.25 (Roache, 1994), r is the refinement factor between from coarse to the fine grid, p is the order of convergence, and $\varepsilon_{i+1,i}$ is the relative error, defined respectively as:

$$p = \frac{\ln\left(\frac{f_3 - f_2}{f_2 - f_1}\right)}{\ln(r)} \quad (5)$$

$$\varepsilon_{i+1,i} = \frac{f_{i+1} - f_i}{f_i} \quad (6)$$

For these simulations, the multiphysics solver described in Section 2 is employed along with the thermophysical properties reported in Table 1. In order to apply initial and boundary conditions consistent with the MSFR operational conditions, a uniform temperature of 890 K and a uniform velocity field along the x direction of -0.8 m/s are imposed to the top of the plug. Furthermore, as boundary conditions, the temperature on the side and bottom of the plug is fixed at 836 K, which corresponds with the initial temperature of the whole freeze plug domain. It is worth noting that these assumptions are only meant to provide insights into freeze valve behaviour while the design prescriptions for such safety components are still being defined.

In order to compare the three meshes, the time at which the mean temperature of the plug reaches steady state is extracted for each simulation. The results and the corresponding number of elements for each mesh are presented in Table 2.

Table 2. Number of elements and time to reach steady state for the three meshes under analysis.

Mesh	Number of elements	Time
(a) Coarse	16 640	5695 s
(b) Intermediate	33 792	4154 s
(c) Fine	67 860	3786 s

Starting from these results, it is possible to evaluate the relative errors, $\varepsilon_{3,2} = 37.10\%$ and $\varepsilon_{2,1} = 9.71\%$, and the corresponding Grid Convergence Index $GCI_{3,2} = 14.52\%$ and $GCI_{2,1} = 3.80\%$, respectively. According to these values, it is possible to appreciate the net improvement moving from the coarsest to the intermediate mesh, whereas the gain is reduced when considering the finest mesh, coherently with an asymptotic approach to convergence. Nevertheless, the multi-physics simulations in this study imply significant computational burden due to the simultaneous

solution of the different physics involved, as well as the need to adopt a refined mesh for the whole domain to properly resolve the molten salt distributions. As a preliminary analysis, this requirement entails the construction of the cylindrical region appended to the lower section of the MSFR adopting the same parameters that characterise the intermediate mesh (b).

4 MSFR analysis

In this section, the overall analysis of the 1/16th symmetric portion of the MSFR primary loop equipped with a region mimicking the freeze plug is proposed. Firstly, the steady state operational conditions for the primary loop are presented, followed by two case studies that differ in the initial and boundary conditions imposed on the freeze valves. The objective is to evaluate the behaviour of the salt plugs under different initial and boundary conditions, taking into account the coupling with the entire fuel loop. These boundary conditions should represent different cooling strategies that could be adopted to guarantee the working state of the valve at steady state. The aim is to evaluate whether the selected geometry of the freeze plug could withstand the harsh MSFR working environment at nominal operating conditions.

4.1 Nominal operating condition analysis

The salt thermophysical properties presented in Table 1 are employed, and the solver is compiled in the OpenFOAM-v9 toolkit (OpenFOAM Official Website, 0000). Furthermore, the parameters characterising the pump and the heat exchanger are tuned to meet an overall thermal power of 187.5 MW, a total volumetric flowrate of $4.5 \text{ m}^3/\text{s}$, and a minimum temperature at the core inlet of 923 K. The mesh domain, along with a close-up on the freeze plug, for a total amount of 1244076 hexahedral elements, is depicted in Fig. 4.

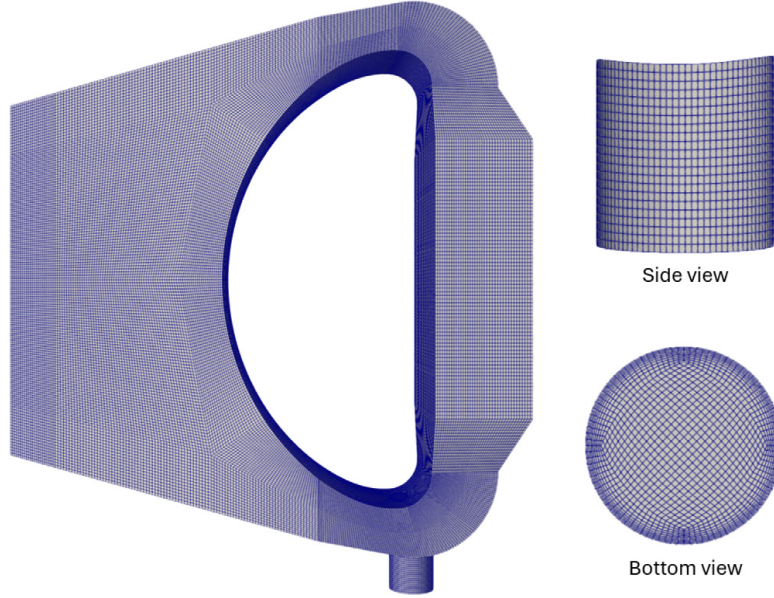


Fig. 4. Mesh domain of the 1/16th of the MSFR primary loop equipped with the freeze plug region. On the right, a close-up on the freeze plug mesh.

This preliminary simulation is necessary to let operational variable distributions stabilise at steady state according to the nominal operating conditions in a mesh that incorporates the cylindrical freeze valve. The need to stabilise the MSFR primary loop distributions is driven by the ultimate goal to isolate the impact of initial and boundary conditions on the freeze valve only in the subsequent simulations. Therefore, the first steps implement a uniform fixed temperature of 836 K for the freeze plug region. According to the solver features presented in Section 2, imposing a temperature below the melting temperature corresponds to a solid region where the liquid fraction is equal to 0 and the velocity field is suppressed to zero, as illustrated in Figs. 5(a), 5(b), and 5(c). As a simplifying assumption in this work, the properties of the salt are kept the same in all the regions despite the different chemical composition envisaged for the freeze plug salt. However, to discard the heat production in the freeze valve region, the neutron fluxes, as well as delayed and decay heat neutron precursor distributions, are forced to zero in the freeze plug domain. This results in a null volumetric power density distribution in the plug domain, as it is possible to appreciate in Fig. 5(d).

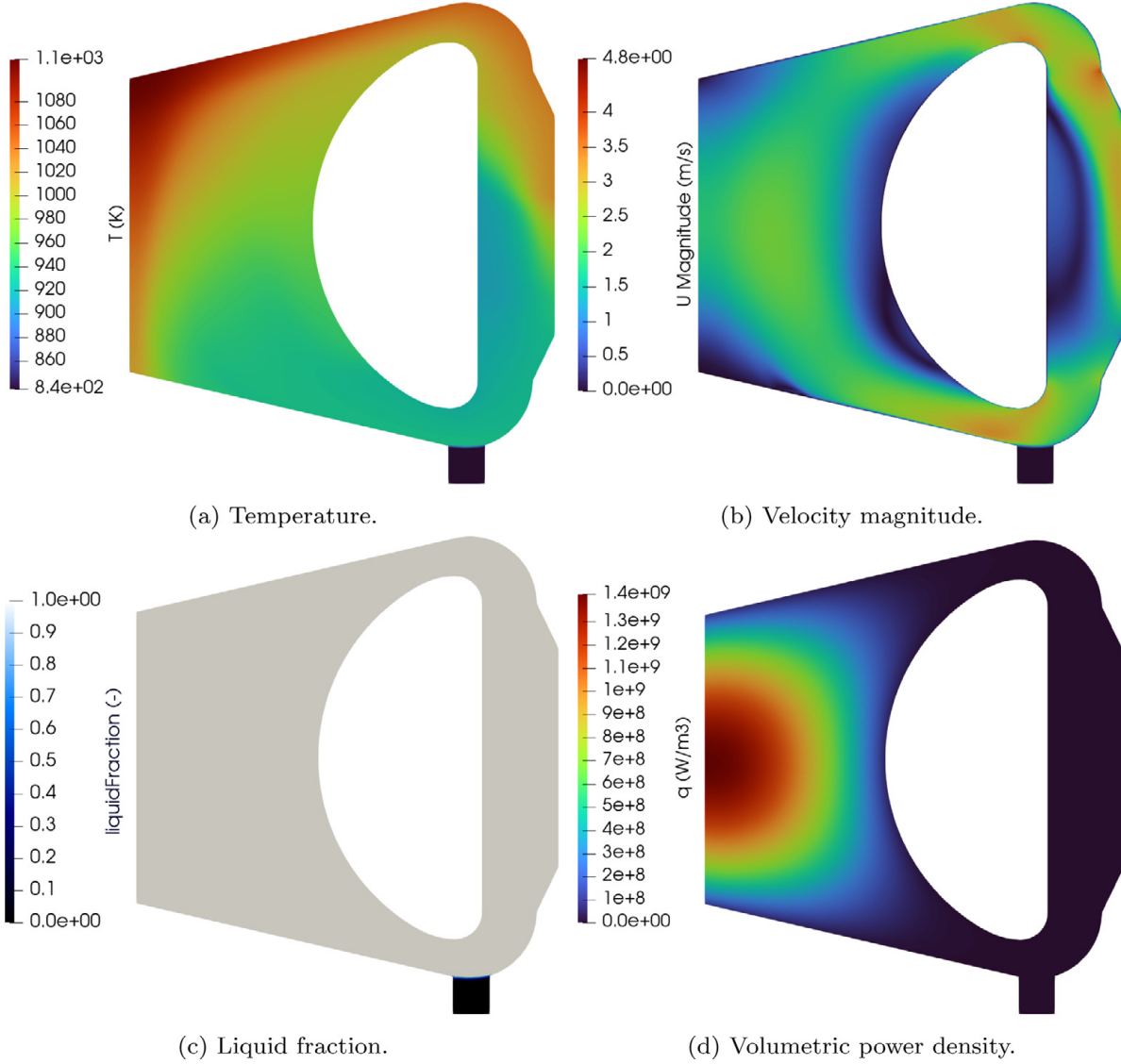


Fig. 5. Steady state distributions for the 1/16th of the MSFR primary loop, equipped with the freeze valve region.

4.2 Case A

The first case study, based on the distributions obtained in Section 4.1, imposes a fixed value temperature boundary condition on the plug side and bottom. Specifically, the wall temperature is set at 836 K, which is also the initial condition for temperature in the freeze valve domain.

4.3 Case B

For the second case study, an ad-hoc boundary condition is implemented based on the external wall heat flux temperature, already available in the OpenFOAM toolkit, but adapted for an incompress-

ible solver. In particular, this boundary condition applies a heat flux condition to temperature on an external wall by imposing the heat transfer coefficient, h , and the ambient temperature, T_a . In this study, $h = 1000 \text{ W/m}^2 \text{ K}$, coherently with other works found in literature (Ilham and Okawa, 2023; Richardson, 1962), and $T_a = 736 \text{ K}$. Furthermore, the initial condition for the freeze plug region is a uniform temperature distribution of 736 K .

4.4 Comparison

Starting from the initial and boundary conditions proposed for *Case A* and *Case B*, summarised in Table 3, the simulations are performed up to 10 s.

Table 3. Initial and boundary conditions for *Case A* and *Case B*.

	<i>Case A</i>	<i>Case B</i>
Side Wall BC	fixedValue 836 K	modifiedExternalWallHeatFluxTemperature $T_a = 736 \text{ K}$, $h = 1000 \text{ W m}^{-2} \text{ K}^{-1}$
Bottom Wall BC	fixedValue 836 K	modifiedExternalWallHeatFluxTemperature $T_a = 736 \text{ K}$, $h = 1000 \text{ W m}^{-2} \text{ K}^{-1}$
Internal Mesh IC	Uniform Value 836 K	Uniform Value 736 K

This time interval is chosen due to the high computational burden of the simulations, but it still allows for a preliminary investigation of the impact of boundary conditions on the freeze valve behaviour. All the plots presented in the following discussion will illustrate the minimum position along the z coordinate where the salt is liquid, referred to as z_{min} . To clarify, in Fig. 6, z_{min} and the reference $z = 0 \text{ m}$ axial coordinate are indicated by the orange arrow superposed to the liquid fraction distribution.

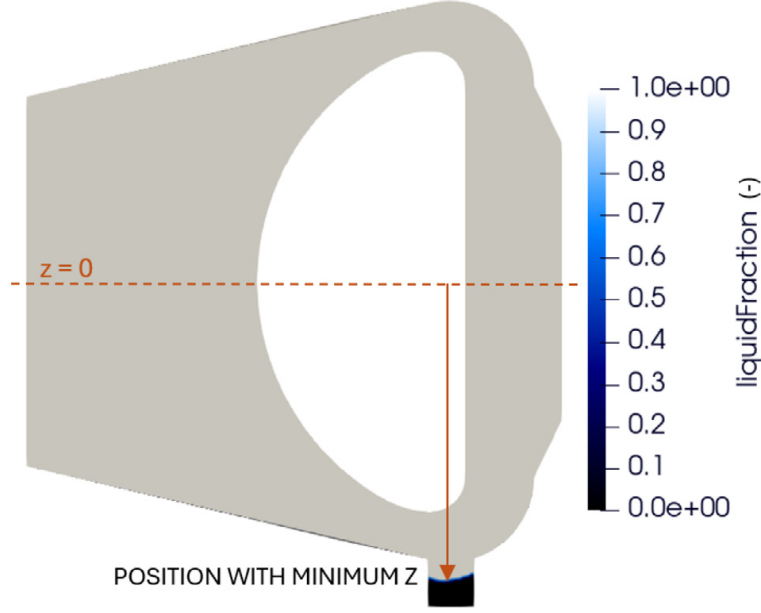


Fig. 6. Liquid fraction distribution, with an arrow starting from the reference $z = 0$ m axial coordinate up to the minimum position along the z coordinate where the salt is liquid (z_{min}).

Firstly, it is worth highlighting the impact of the *mushy zone* on the simulation results. Fig. 7 illustrates a comparison of the minimum melting front position, z_{min} , considering two different liquid fraction thresholds, i.e., 0.2 and 0.5. *Case A* is chosen as representative but the same results hold for *Case B*. In both scenarios, the difference extends to 5.5 mm by the end of the simulation. This value highlights the impact of the *mushy zone* modelling approach, which does not capture the melting interface but instead smears it out over a finite range, establishing a relation between the melting front characterisation and the selected liquid fraction threshold. As already discussed in previous works (Tartaglia et al., 2023), properly selecting a priori the temperature width of the *mushy zone* around melting temperature is fundamental to prevent numerical issues in the results.

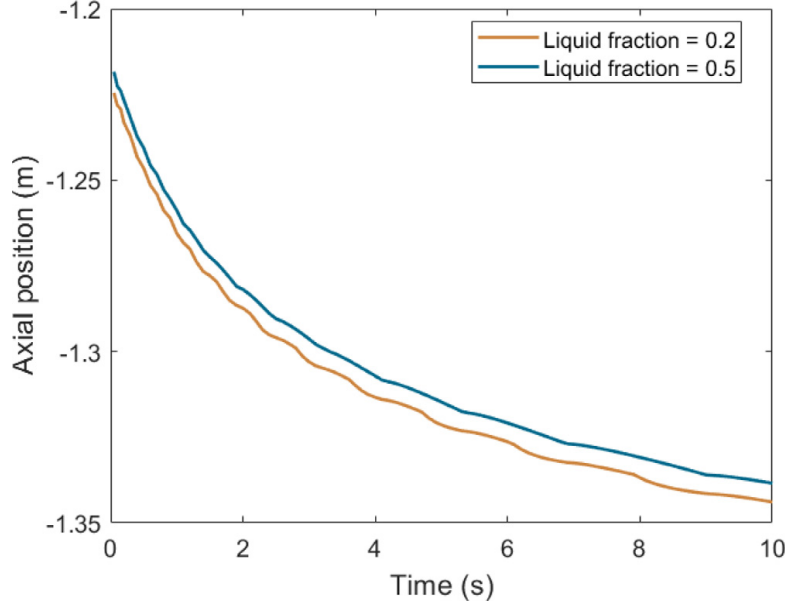


Fig. 7. Comparison of the minimum position along the z coordinate where the salt is liquid when considering a threshold of 0.2 or 0.5 for the liquid fraction, for *Case A*.

Figs. 8 and 9 show the evolution of the freeze plug solid domain during the simulation at different time steps for *Case A* and *Case B*, respectively. First, the impact of convective mechanisms on the melting of the salt is emphasised by the preferential direction of the salt melting front in both cases, which is enhanced along the salt motion above the plug. Furthermore, for *Case A*, in the first layer of cells on the plug side, the salt remains solid due to the fixed temperature boundary condition below the melting temperature. In fact, when active cooling boundary condition is considered, such as *Case B*, imposing a heat transfer coefficient clearly indicates that the salt adherent to the side of the plug melts as well. Secondly, the combined effect of a reduced initial temperature in the freeze valve, along with a heat sink characterised by a heat transfer coefficient and an external temperature, significantly slows down the melting of the plug. To highlight this delay, the minimum position along the z coordinate where the salt is liquid is plotted for the two case studies in Fig. 10.

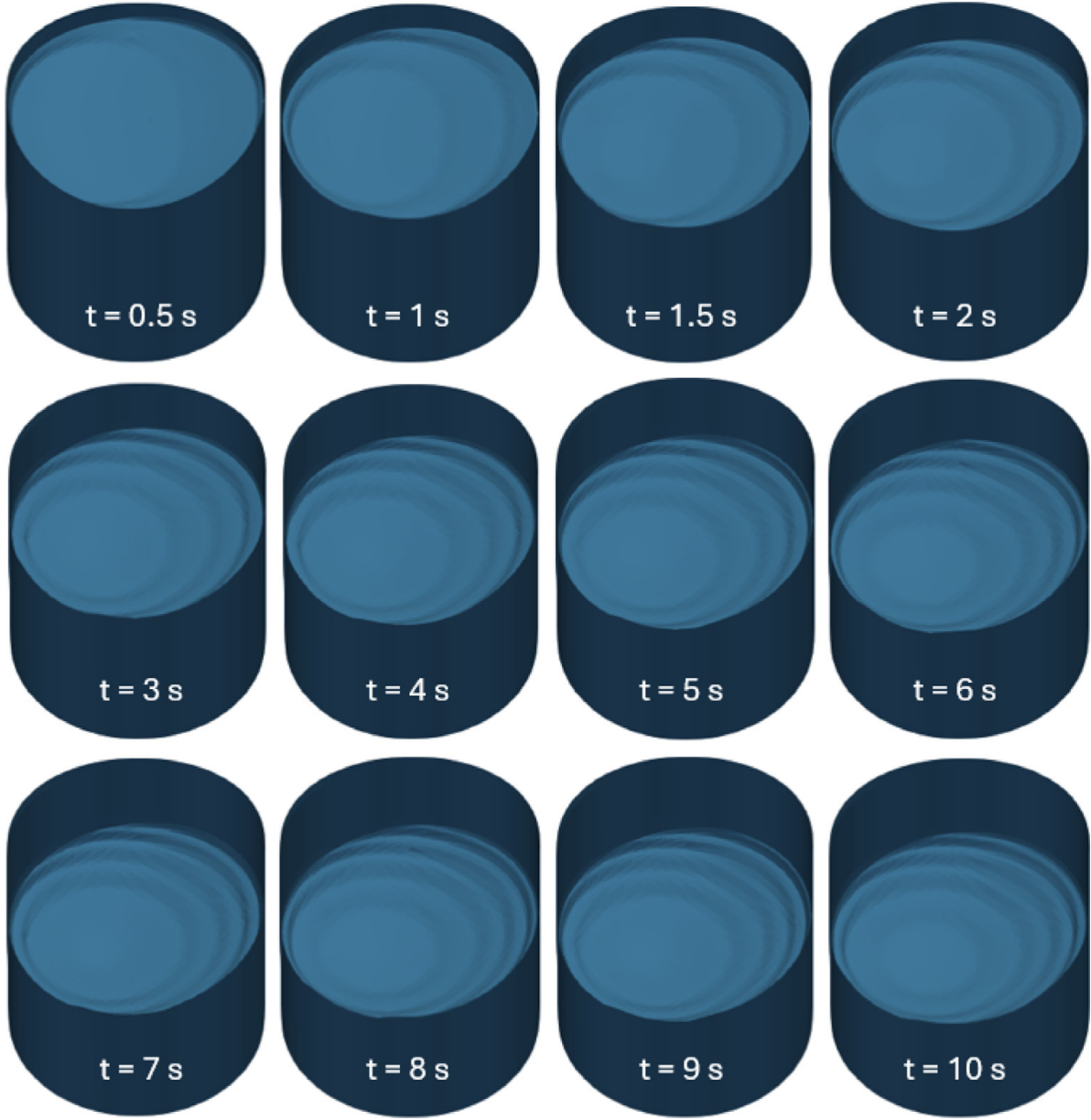


Fig. 8. Evolution of the solid domain in the freeze plug for *Case A*.

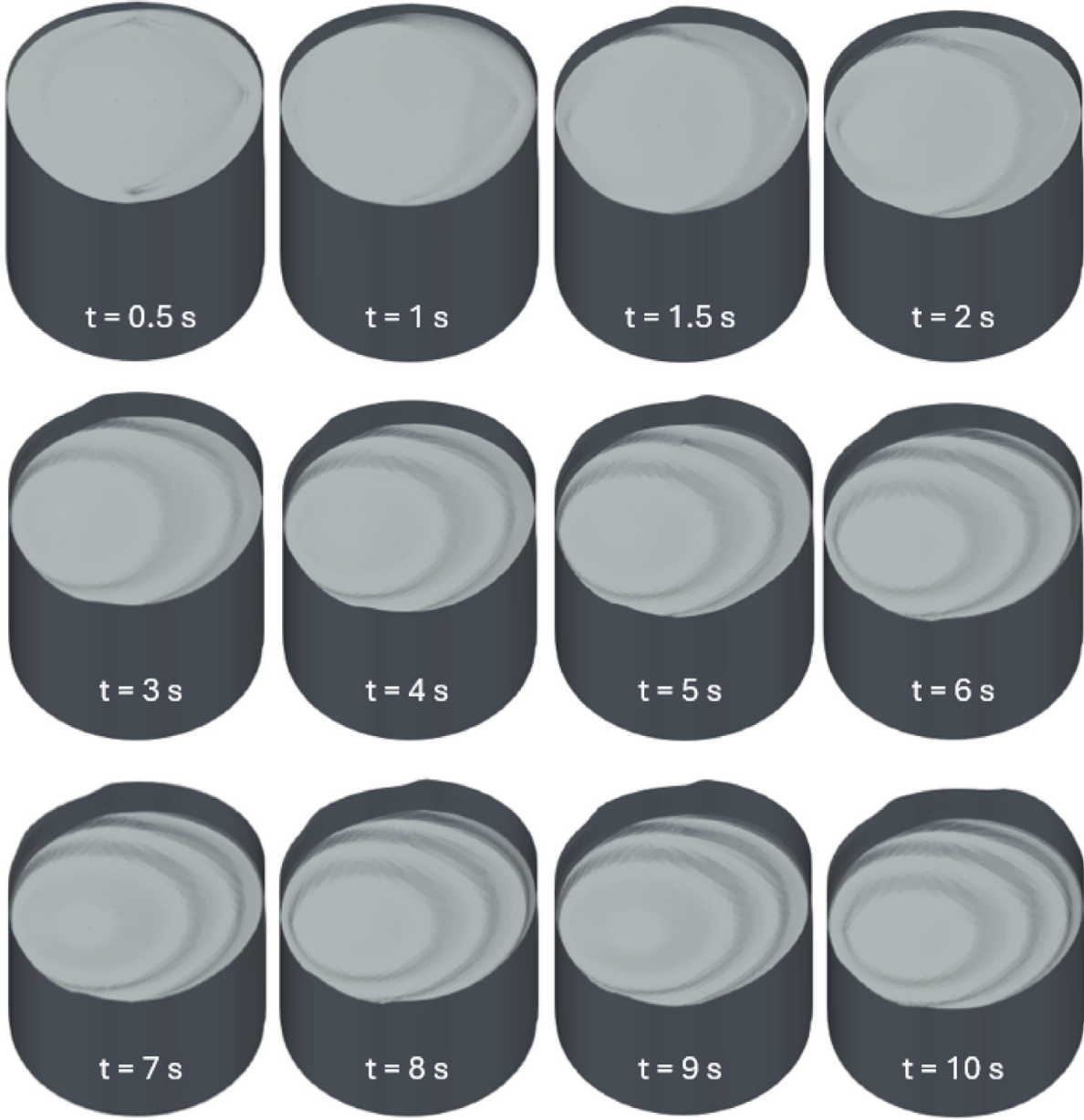


Fig. 9. Evolution of the solid domain in the freeze plug for *Case B*.

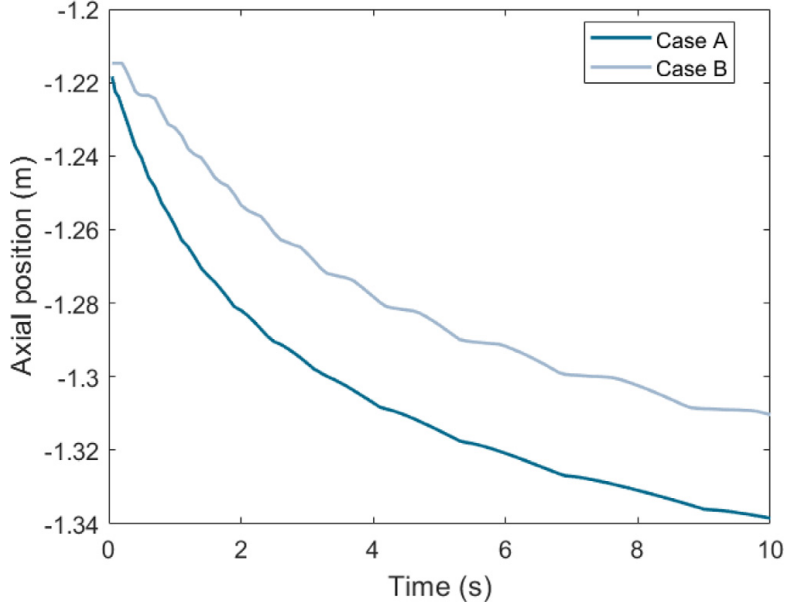


Fig. 10. Comparison of the time profiles of the minimum position along the z coordinate where the salt is liquid between *Case A* and *Case B*.

In this figure, it is possible to observe oscillations characterised by progressively wider plateaux, particularly evident in *Case B*. To further investigate this trend, Fig. 11 illustrates the liquid fraction distribution on a vertical slice at different time instants for *Case B*. A threshold value of 0.5 is adopted for the liquid fraction to identify the liquid front, as for Fig. 10. The first observation is that the progression of the liquid front exhibits distinct axial and radial dynamics. Specifically, when the front undergoes a slight axial advancement, the surrounding salt at the corresponding axial level melts radially in a few subsequent time instants. This behaviour is outlined by the nearly horizontal segment of the melting front observed at 4.5 s in Fig. 11 prior to the axial advancement in depth of the melting front. As a result, a purely axial representation of the liquid front progression with z_{min} lacks part of the information, although it still reflects the overall melting behaviour of the salt plug. From a numerical standpoint, this trend with plateaux can primarily be attributed to the *mushy zone* modelling approach, such that the liquid front is inherently distributed over a range of values. Consequently, the graphical representation requires the definition of a threshold value for the liquid fraction since the liquid front is not resolved. Furthermore, the *mushy zone* tends to extend preferentially in the radial direction before the melting front advances to the next axial cell. This behaviour results in a relatively constant value of z_{min} for a few instants, with the threshold value set to 0.5. In principle, these numerical artefacts could be mitigated by adopting a significantly finer axial mesh resolution and by minimising the width of the *mushy zone*.

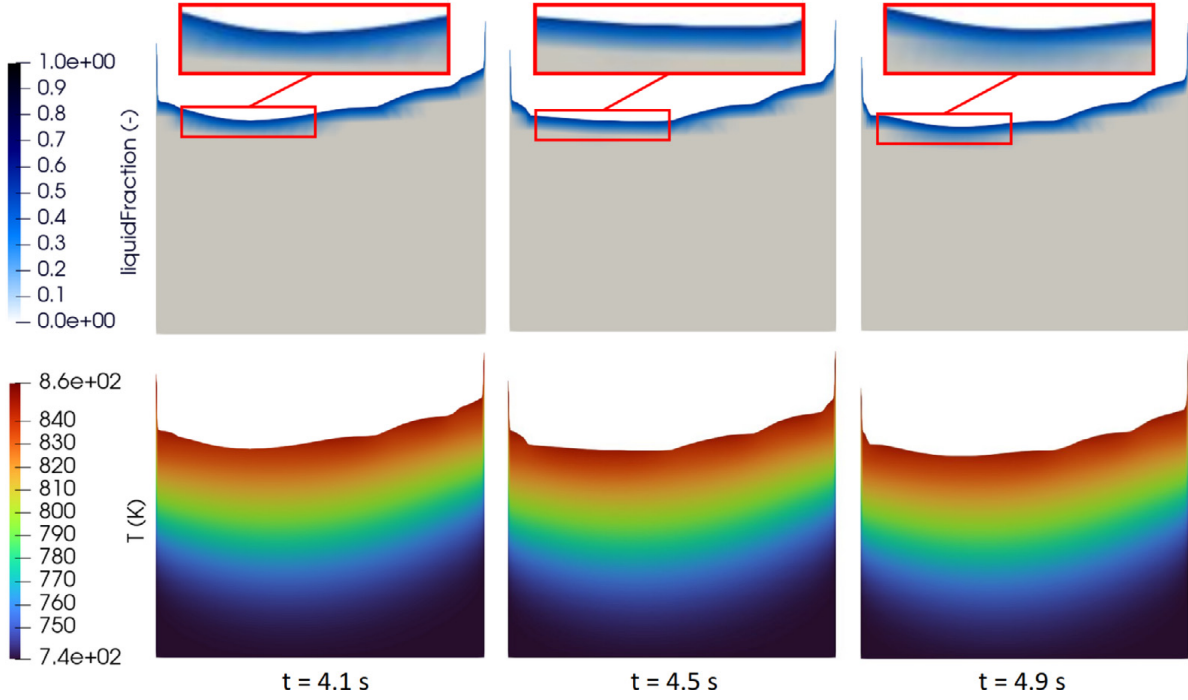


Fig. 11. Temperature (bottom) and liquid fraction (top) distributions at different time instants on a vertical slice for the freeze valve in *Case B*. The threshold is set to a liquid fraction of 0.5.

With the aim of obtaining a preliminary estimate of the melting time of the whole freeze plug region, the profiles illustrated are fitted thanks to exponential functions. Then, the melting time is extracted as the time at which z_{min} reaches -1.41 m, i.e., the bottom of the plug. The fitting curve for both scenarios is exponential, in the form of:

$$f(x) = a \cdot \exp(b \cdot x) + c \cdot \exp(d \cdot x) \quad (7)$$

where a , b , c , and d are the fitting coefficients, presented for the two cases in Table 4.

Table 4. Coefficients for the exponential fitting curves of the z_{min} time profiles for *Case A* and *Case B*.

Coefficient	<i>Case A</i>	<i>Case B</i>
(a)	-1.298 m	-1.299 m
(b)	$3.213 \cdot 10^{-3} \text{ s}^{-1}$	$1.261 \cdot 10^{-3} \text{ s}^{-1}$
(c)	$8.005 \cdot 10^{-2} \text{ m}$	$8.902 \cdot 10^{-2} \text{ m}$
(d)	$-6.065 \cdot 10^{-1} \text{ s}^{-1}$	$-2.848 \cdot 10^{-1} \text{ s}^{-1}$

This procedure leads to a melting time of 25.5 s for *Case A* and 64.4 s for *Case B*, underlining the importance of selecting appropriate boundary and beginning conditions to accurately predict frozen plug behaviour. Indeed, the ultimate objective of the preliminary analysis proposed in this

paper is to investigate the feasibility of freeze valves as passive safety systems. Given that even at initial temperature only 5 K below melting, the first estimate for melting time is in the order of tens of seconds, the role and applicability of freeze plugs as passive safety barriers should be carefully assessed. As a further comment, *Case B* demonstrates that, despite the implementation of active cooling heat transfer capability, the plug is expected to melt completely. As a result, it is critical that the geometry and position of the plug are designed to guarantee the complete closure of the valve at steady state. Overall, it is possible to state that the design prescriptions and, in turn, the initial and boundary conditions imposed on the domain need further investigation before assessing a more accurate estimate of the melting time. In any case, the model presented in this work proved its flexibility to give indications on the freeze valve behaviour under different working conditions. Furthermore, it is possible to highlight that a key advantage of employing a 3D fully coupled MSFR model, which includes the freeze valve, consists in the direct evaluation of boundary conditions arising from the interaction between the primary circuit salt flow and the freeze plug salt. This approach captures the local variations in temperature and velocity profiles along the frozen solid surface of the plug, thereby avoiding the simplification of imposing spatially uniform conditions on a flat boundary surface.

5 Conclusions

This paper proposed a preliminary analysis of the freeze valve behaviour, which is currently under consideration as a safety barrier for the MSFR and is designed to melt in the event of an unwanted temperature rise. The mechanism relying on the state of the plug aligns with the approach of passive safety, justifying the attractiveness of deepening the knowledge of the behaviour of freeze valves. The proposed solver incorporates melting and solidification phenomena within a multi-physics solver, coupling neutronics and thermal-hydraulics, by modelling the latent heat and the velocity transition. The domain consists of a 3D representation of 1/16th of the MSFR fuel loop.

Firstly, a sensitivity analysis allowed to assess the impact of mesh refinement on the cylindrical zone added to the MSFR primary loop to mimic the freeze plug. To lighten the computational burden, the meshes were tested on a square cylinder of 20 cm diameter, with boundary conditions representing the coupling with the MSFR primary circuit. The GCI and the relative error were calculated based on the time to reach steady state for three meshes of a square cylinder with different refinement levels. Refining the mesh leads to reduced GCI and relative error, so the approach to convergence is satisfied. Moreover, due to the computational burden of the simulations, the intermediate refinement mesh is chosen. Therefore, a cylindrical domain is added below the cold leg of the primary loop with the same parameters employed for the intermediate refinement mesh. Then, a steady-state simulation is performed to let the salt distributions stabilise while forcing the

freeze plug in a solid state. The results serve as starting points for two case scenarios. Indeed, different boundary and initial conditions are imposed on the freeze plug region in order to assess the influence of the working conditions on the melting time of the freeze valve. With the aim to investigate different cooling strategies for the valve, fixed value or external wall heat flux boundary conditions are imposed to the plug walls. The objective is to verify whether this geometry can maintain the freeze plug in a working state, ensuring the complete closure while withstanding the MSFR nominal operating conditions at steady state. First, *Case A* involves a uniform fixed temperature distribution of 836 K as initial and boundary conditions. In this scenario, the fact that the plug melting time is estimated to be approximately 25 s, even when the salt starts from 5 K below melting temperature, raises questions about the applicability in safety scenarios, requiring further investigation. On the other hand, when considering active cooling boundary conditions and an initial temperature of 736 K, the 20 cm of salt are expected to melt completely in almost 65 s. The predicted complete melting of the plug emphasises the importance of properly designing the geometry and position of the freeze plug to guarantee correct functioning in nominal operation. In both cases, the convective contribution to melting phenomena is evident from the structure of the melting front, which demonstrates a preferential direction along the salt flow motion.

The goal of this study was to conduct preliminary investigations into freeze plug behaviour employing a domain that included both the MSFR primary circuit and the freeze valve. The initial and boundary conditions have a significant impact on the melting time of the plug, requiring more accurate design prescriptions to assess its applicability as a passive safety barrier in an MSFR. In any case, the modelling framework demonstrated its flexibility in assessing freeze plug response under various working conditions and eventually transient scenarios.

CRedit authorship contribution statement

S. Deanesi: Writing – original draft, Investigation, Formal analysis. **A. Cammi:** Supervision, Conceptualization. **S. Lorenzi:** Supervision, Investigation, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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